

# Proyecto Final Econometría II

En esta seccion se realiza un pequeño preprocesamiento de los datos apra tener un DF de pandas para trabajar.

|     | Fecha      | Tasa_participacion |
|-----|------------|--------------------|
| 4   | 2005-01-01 | 40.351692          |
| 5   | 2005-02-01 | 40.432370          |
| 6   | 2005-03-01 | 41.179709          |
| 7   | 2005-04-01 | 39.732705          |
| 8   | 2005-05-01 | 41.027773          |
| ... | ...        | ...                |
| 253 | NaT        | NaN                |
| 254 | NaT        | NaN                |
| 255 | NaT        | NaN                |
| 256 | NaT        | NaN                |
| 257 | NaT        | NaN                |

254 rows × 2 columns

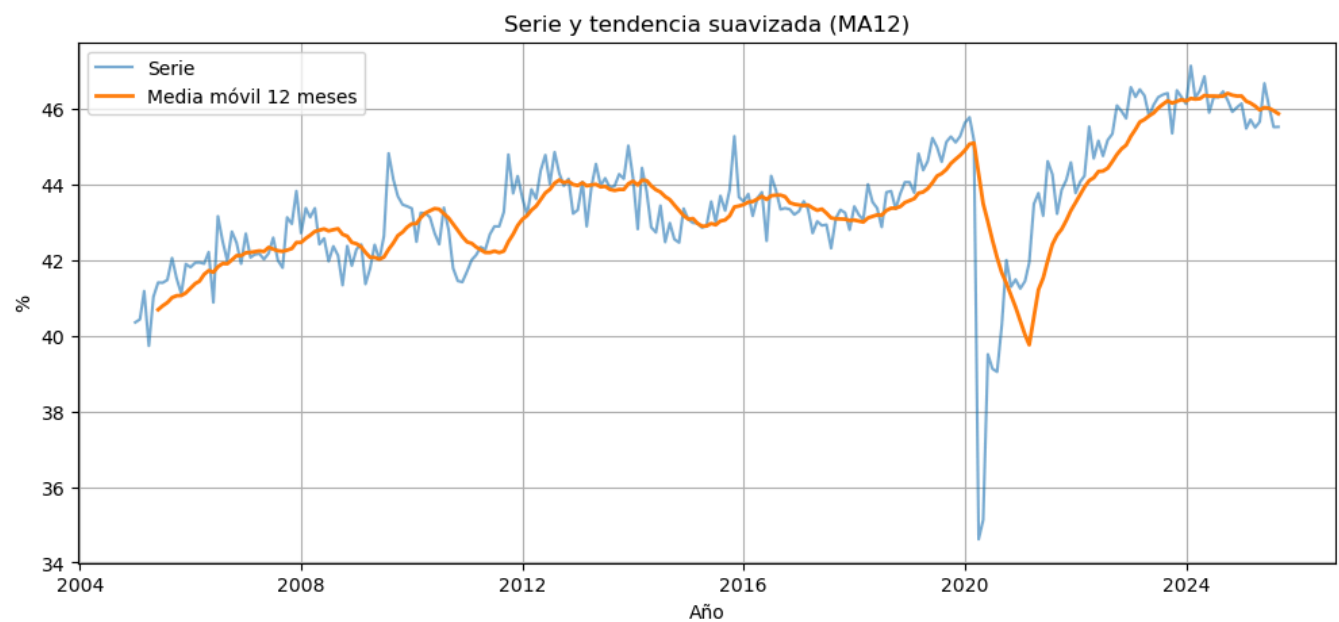
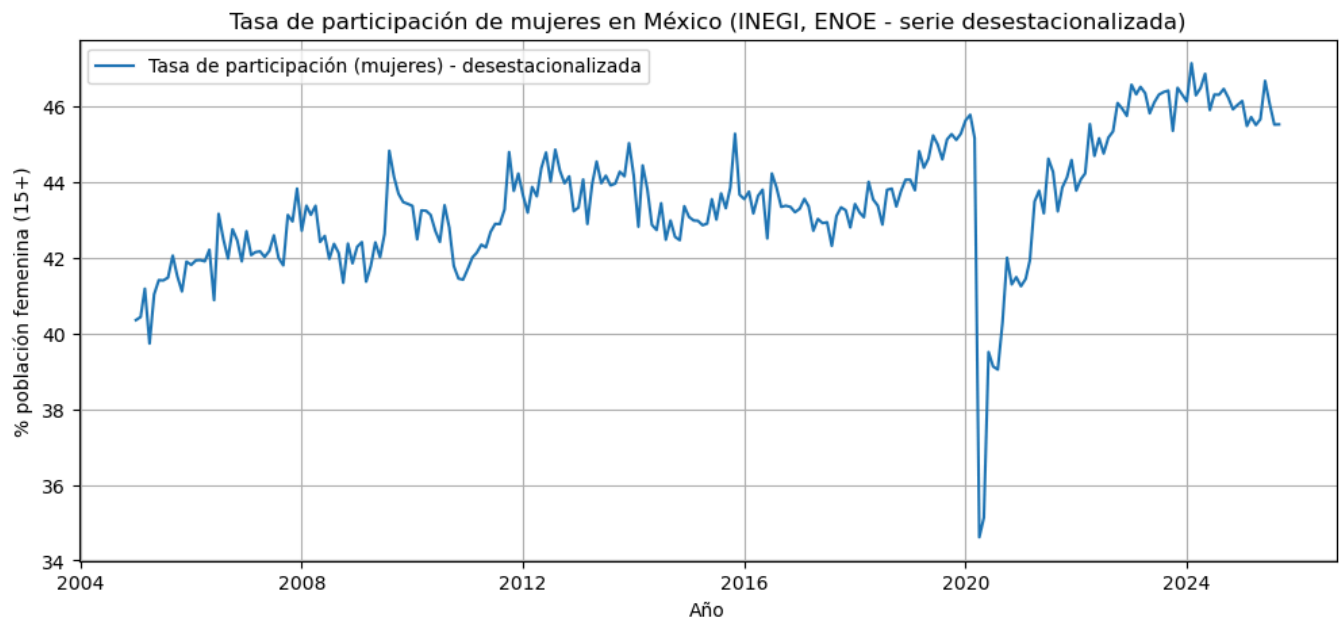
Comienza la exploración de los datos.

Periodo detectado: 2005-01 a 2025-09

Resumen estadístico:

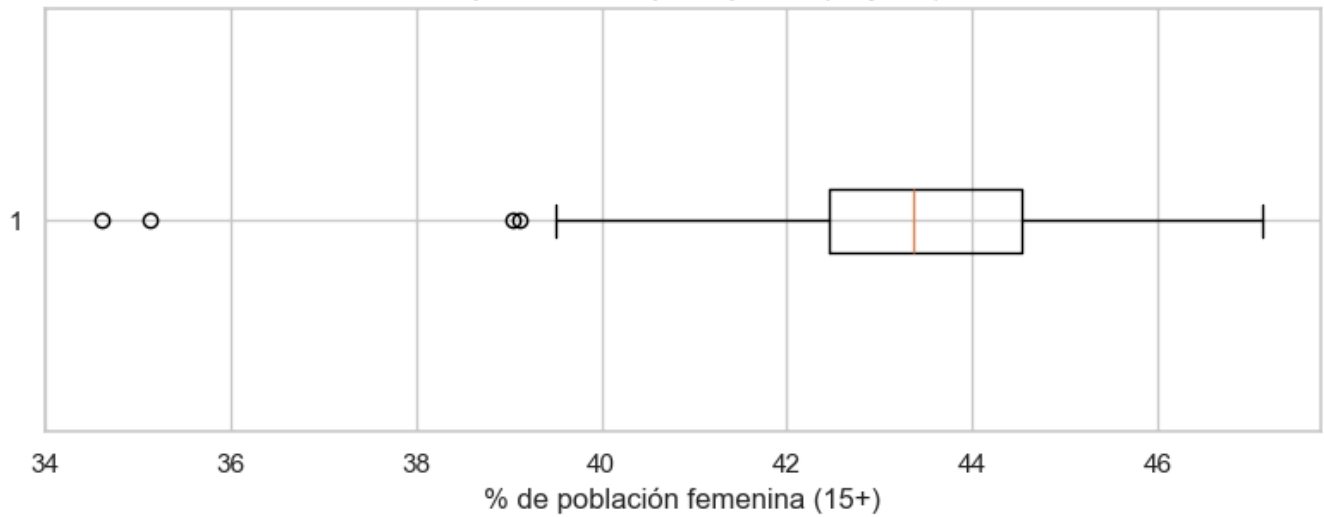
|       |            |
|-------|------------|
| count | 249.000000 |
| mean  | 43.472531  |
| std   | 1.748154   |
| min   | 34.616667  |
| 25%   | 42.457499  |
| 50%   | 43.366709  |
| 75%   | 44.539689  |
| max   | 47.135221  |

Archivo limpio guardado como: tasa\_participacion\_mujeres\_clean.csv

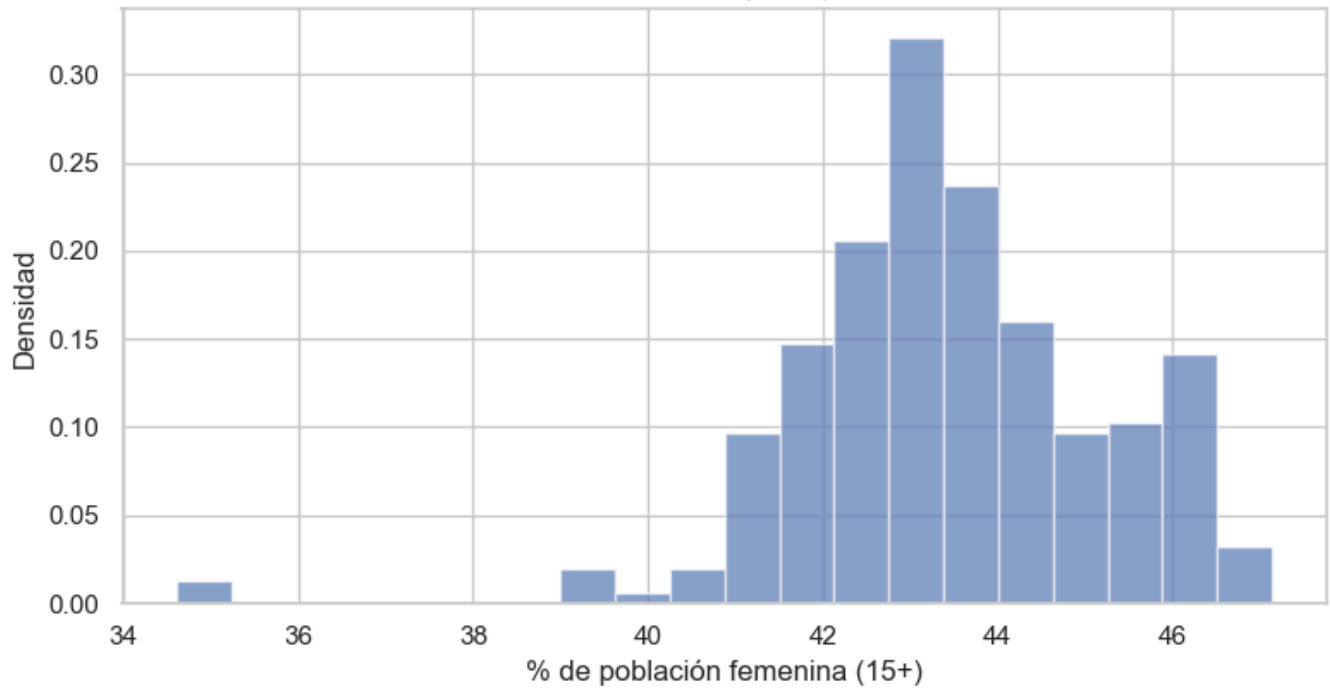


## Exploración y Transformaciones

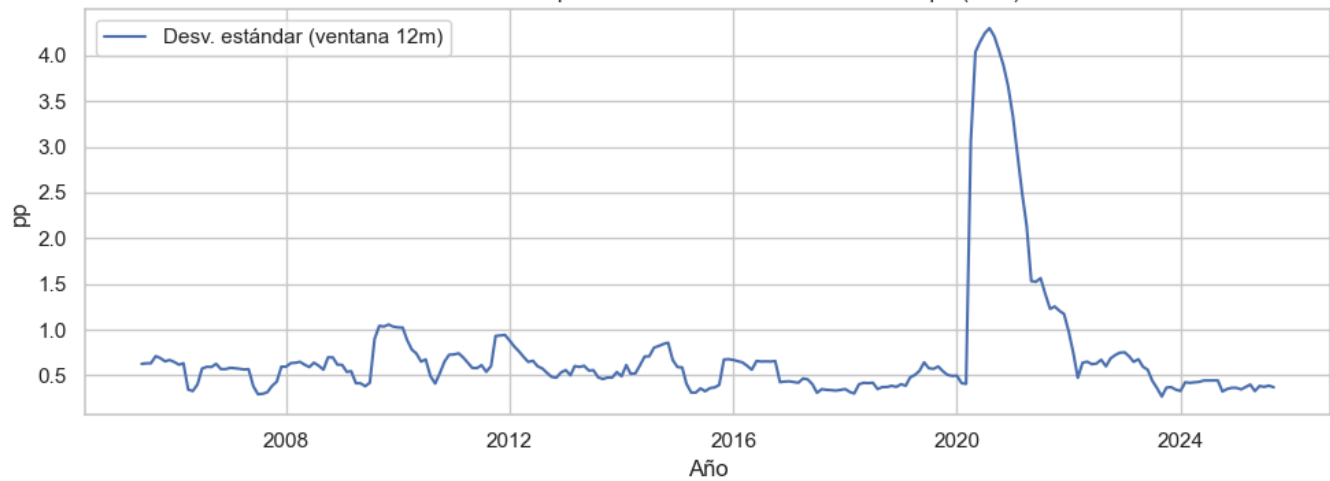
Boxplot - Tasa de participación (mujeres)

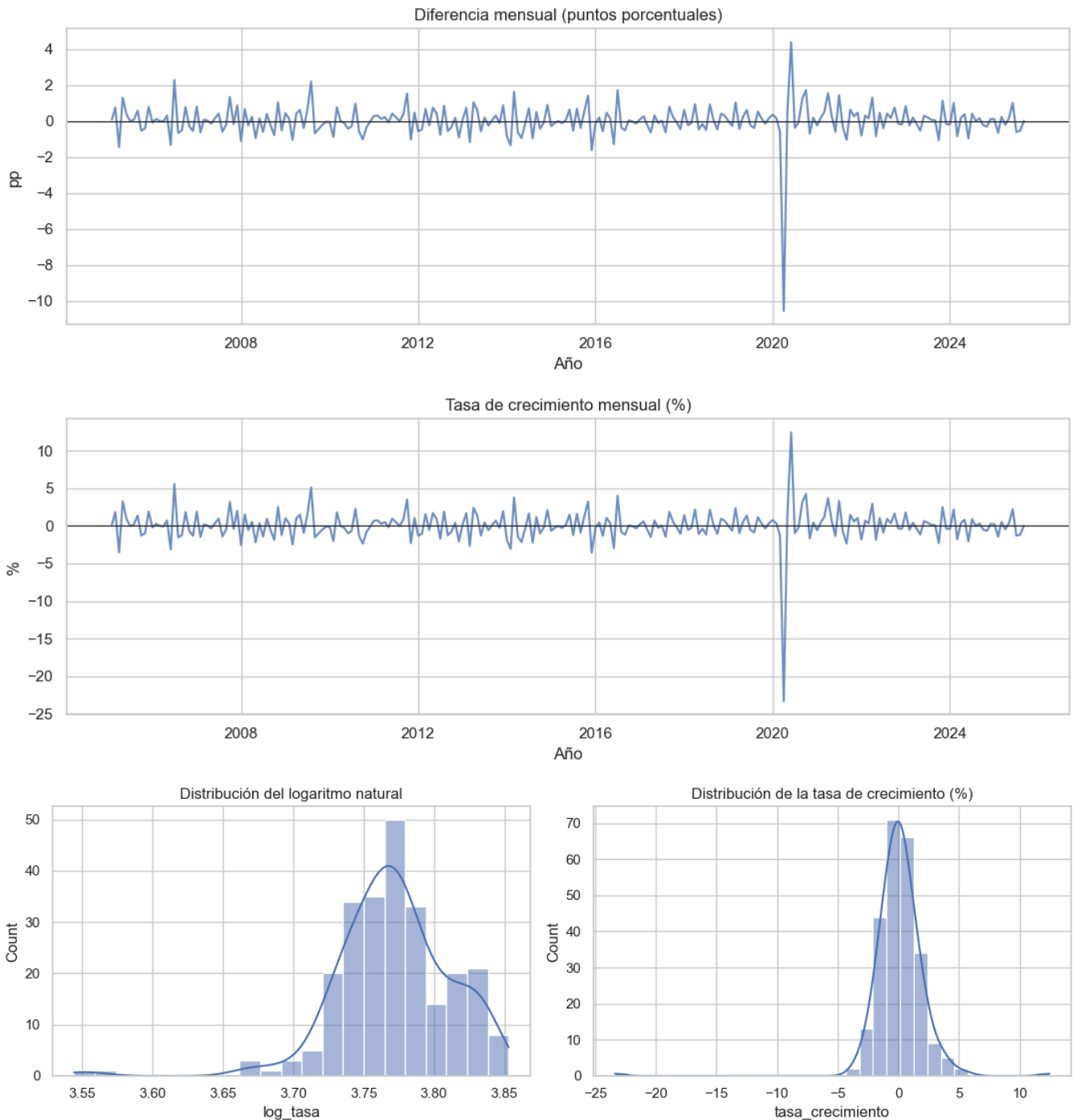


Distribución de la tasa de participación femenina



Variabilidad temporal: desviación estándar en el tiempo (12m)





## (Sección 2) Estadísticos y exploración

Periodo analizado: 2005-01 – 2025-09 | Observaciones: 249

Media: 43.47 %, Desv.Est.: 1.75 pp, Min: 34.62 %, Mediana: 43.37 %, Max: 47.14 %

Asimetría (skew): -0.819 | Curtosis (Fisher): 3.619

Posibles outliers (IQR): 4 observación(es) (ver meses listados abajo).

Meses marcado(s) como posible(s) outlier por IQR:

Fecha

2020-04-01 34.616667

2020-05-01 35.132391

2020-07-01 39.125171

2020-08-01 39.044937

# Verificación de estacionariedad

Prueba Dickey-Fuller Aumentada (ADF) - Serie en niveles

ADF Statistic: -3.1322314769472266

p-value: 0.02425805585029728

# Lags Used: 2

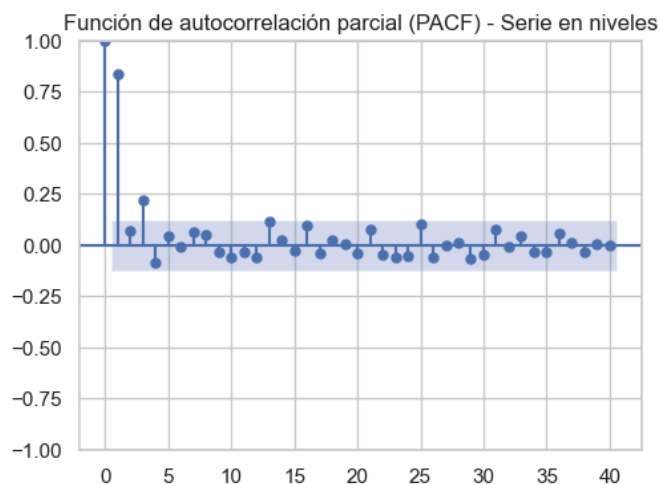
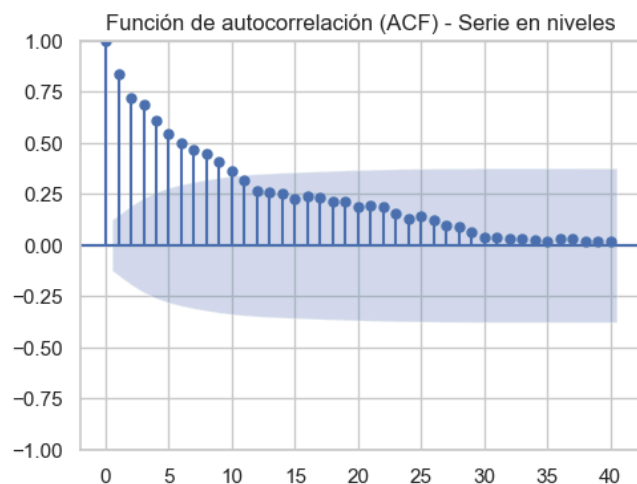
# Observations: 246

Critical Value 1%: -3.457215237265747

Critical Value 5%: -2.873361841566324

Critical Value 10%: -2.5730700760129555

Conclusión: Rechazar  $H_0$  (serie estacionaria)



Prueba Dickey-Fuller Aumentada (ADF) - Primera diferencia

ADF Statistic: -15.902705355748816

p-value: 8.293961936443366e-29

# Lags Used: 1

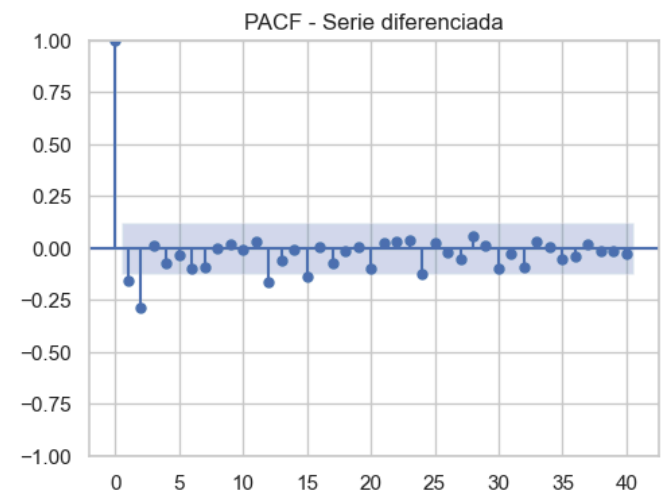
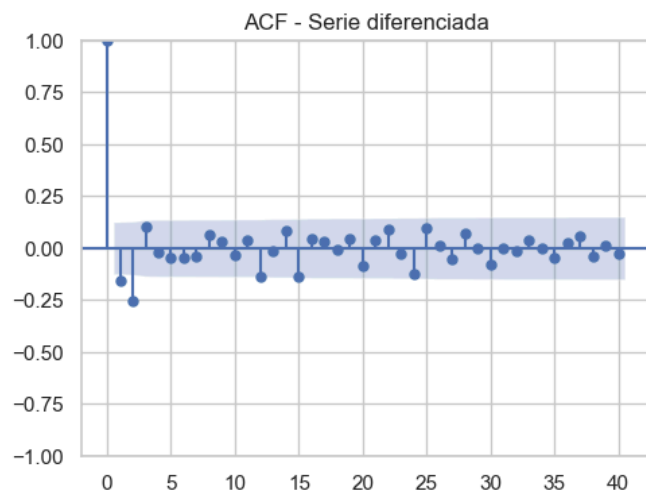
# Observations: 246

Critical Value 1%: -3.457215237265747

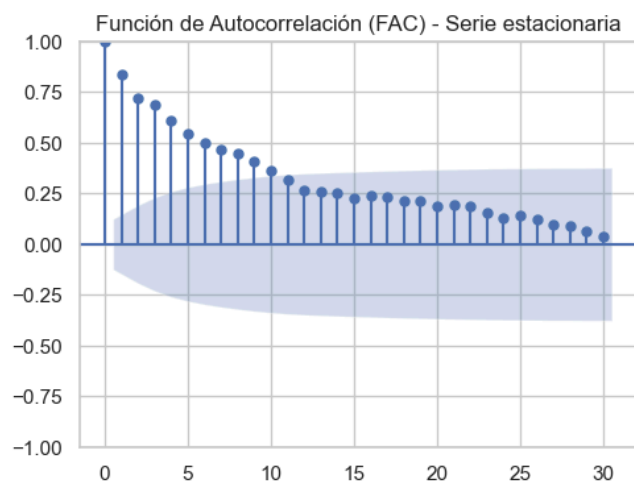
Critical Value 5%: -2.873361841566324

Critical Value 10%: -2.5730700760129555

Conclusión: Rechazar  $H_0$  (serie estacionaria)

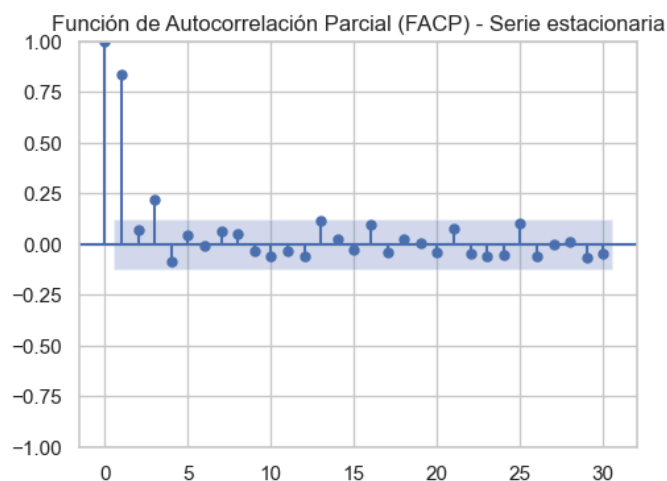


## PARTE 4: Diagnóstico con FAC y FACP



Primeros coeficientes de la FAC:

Lag 0: 1.000  
Lag 1: 0.836  
Lag 2: 0.720  
Lag 3: 0.684  
Lag 4: 0.608  
Lag 5: 0.546  
Lag 6: 0.500  
Lag 7: 0.468  
Lag 8: 0.449  
Lag 9: 0.411  
Lag 10: 0.360



Primeros coeficientes de la FACP:

Lag 0: 1.000  
Lag 1: 0.839  
Lag 2: 0.072  
Lag 3: 0.227  
Lag 4: -0.086  
Lag 5: 0.045  
Lag 6: -0.011  
Lag 7: 0.070  
Lag 8: 0.052  
Lag 9: -0.033  
Lag 10: -0.062

## Estimación del modelo

AR(1)

=== Modelo AR(1) estimado sobre la serie en niveles ===

#### SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(1, 0, 0)        Log Likelihood      -337.648
Date:             Wed, 19 Nov 2025      AIC                 681.296
Time:             22:57:58              BIC                 691.848
Sample:           01-01-2005            HQIC                685.543
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
const         43.4494      0.506     85.895      0.000     42.458     44.441
ar.L1          0.8480      0.031     27.307      0.000      0.787      0.909
sigma2         0.8773      0.024     36.911      0.000      0.831      0.924
=====
Ljung-Box (L1) (Q):           1.42    Jarque-Bera (JB):           34836.33
Prob(Q):                     0.23    Prob(JB):                  0.00
Heteroskedasticity (H):       3.80    Skew:                      -5.10
Prob(H) (two-sided):          0.00    Kurtosis:                   60.04
=====
```

#### Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

#### Parámetros principales:

Constante: 43.4494

Phi\_1 (AR1): 0.8480

AIC: 681.30

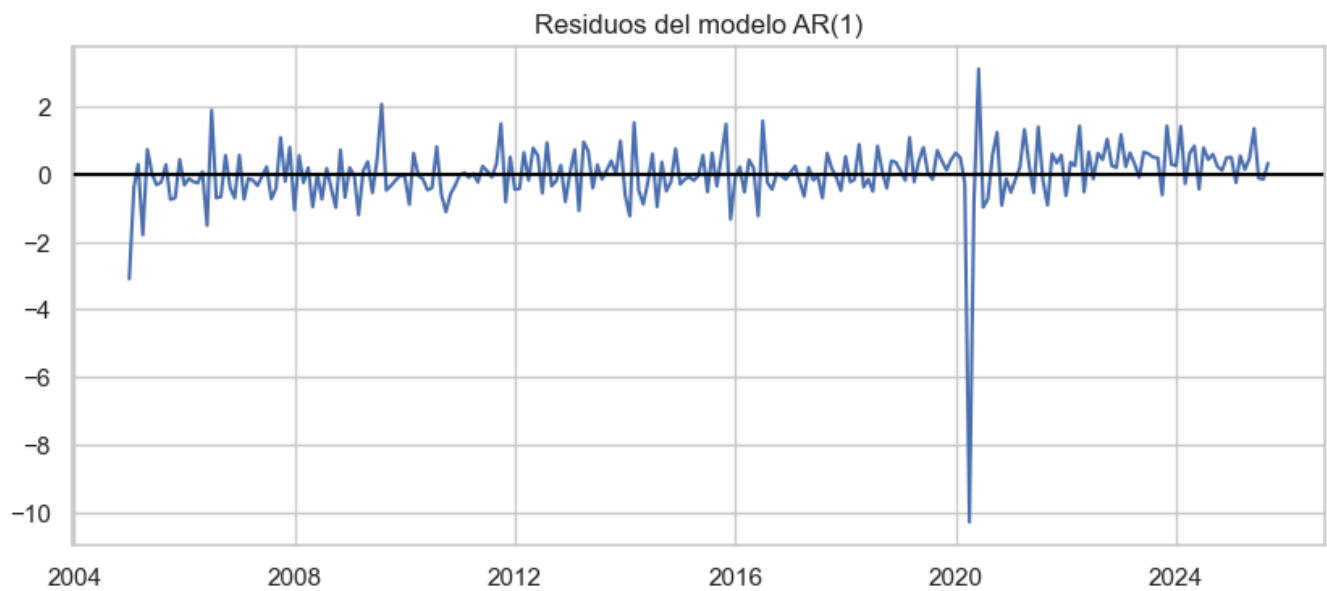
BIC: 691.85

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
```

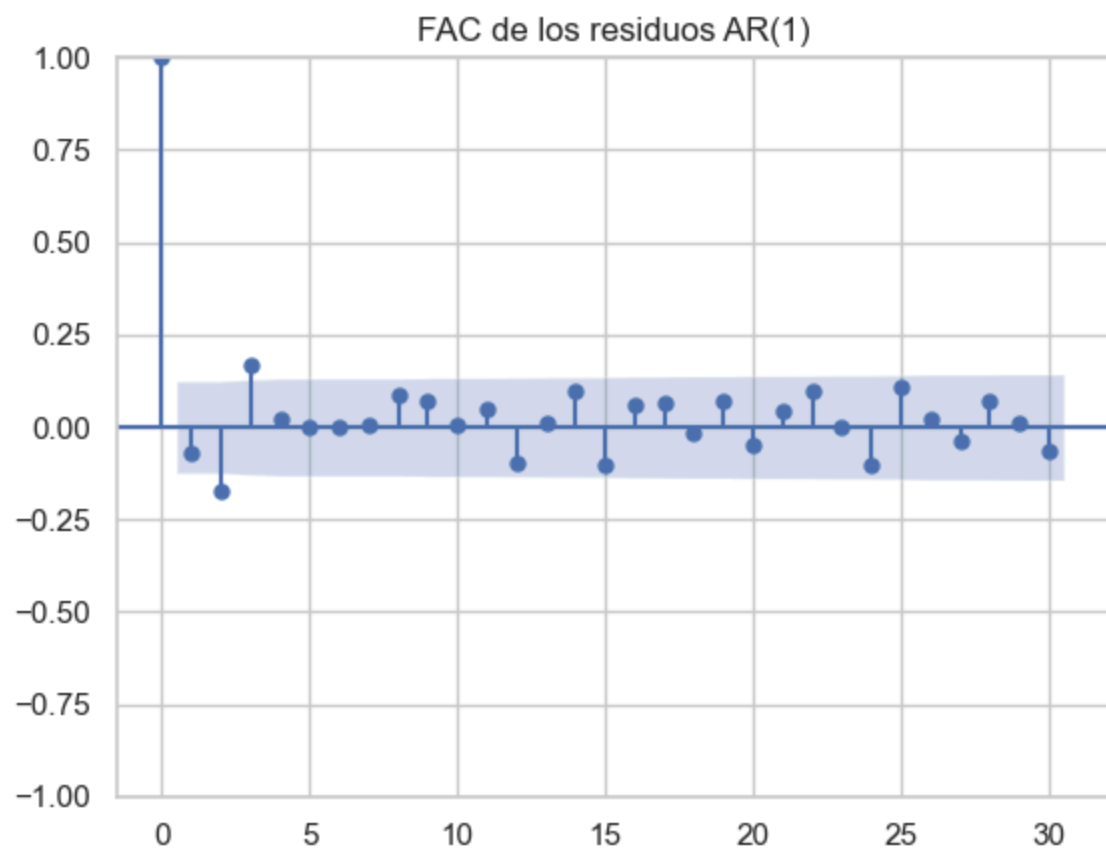
=== Prueba de Ljung-Box (residuos AR(1)) ===

```
    lb_stat  lb_pvalue
10  19.270314  0.036961
```

Conclusión: Se rechaza  $H_0$ ; los residuos muestran autocorrelación (modelo AR(1) puede ser insuficiente).

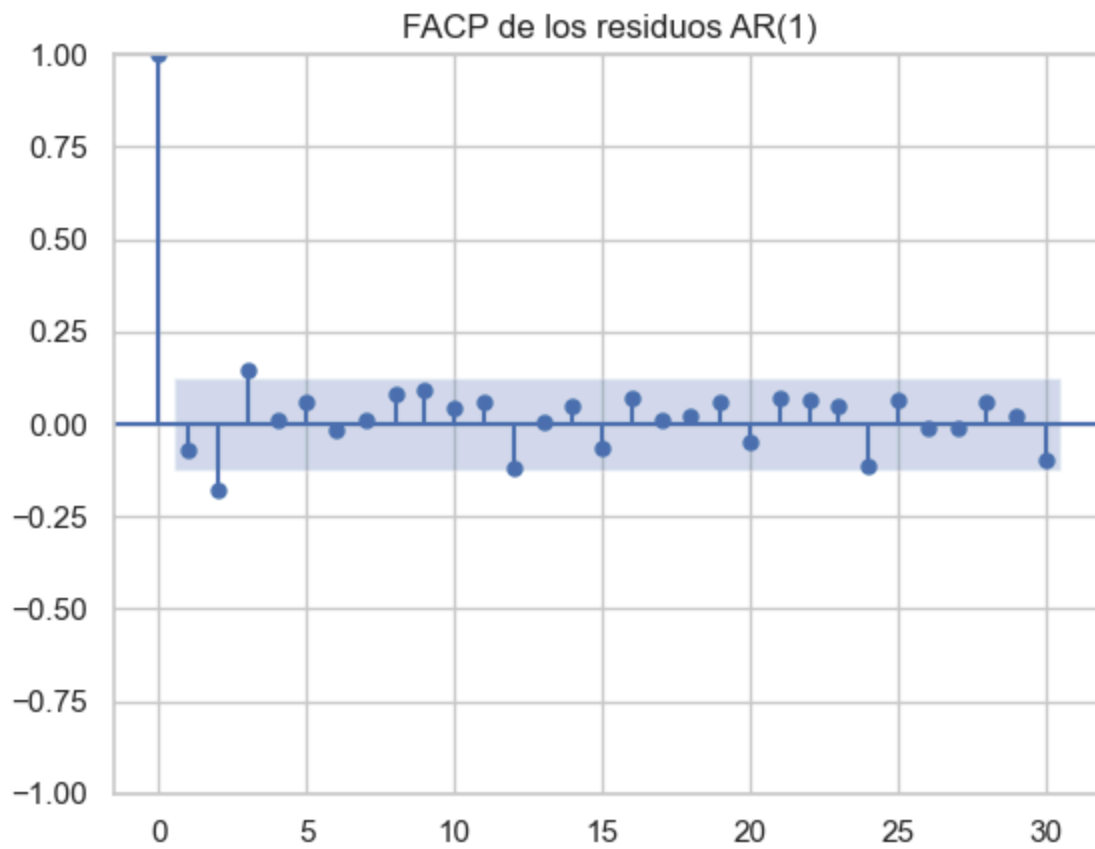


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MA(1)

=== Modelo MA(1) estimado sobre la serie en niveles ===

#### SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(0, 0, 1)        Log Likelihood      -399.829
Date:            Wed, 19 Nov 2025       AIC                 805.658
Time:            22:57:59               BIC                 816.211
Sample:          01-01-2005             HQIC                809.906
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
const         43.4728     0.151    288.085     0.000     43.177     43.769
ma.L1          0.7455     0.039     19.339     0.000      0.670      0.821
sigma2         1.4482     0.053     27.331     0.000      1.344      1.552
=====
Ljung-Box (L1) (Q):          33.40    Jarque-Bera (JB):          2056.62
Prob(Q):                    0.00    Prob(JB):              0.00
Heteroskedasticity (H):      2.81    Skew:                 -1.87
Prob(H) (two-sided):         0.00    Kurtosis:             16.57
=====
```

#### Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

#### Parámetros principales:

Constante: 43.4728

Theta\_1 (MA1): 0.7455

AIC: 805.66

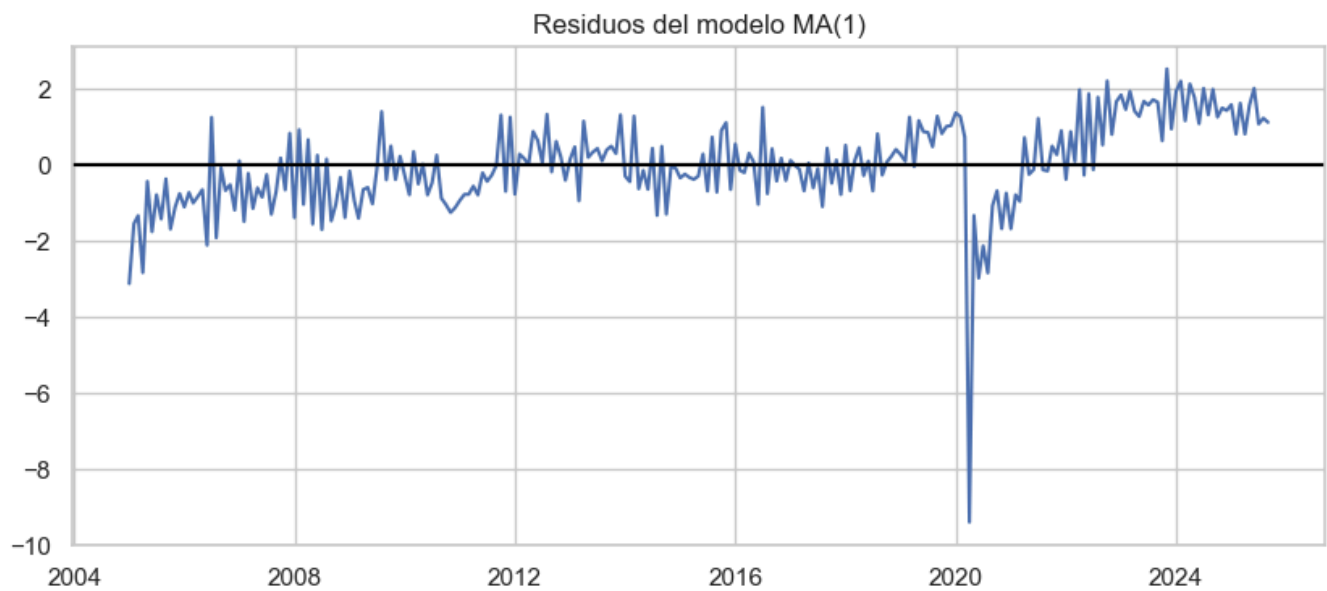
BIC: 816.21

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
    self._init_dates(dates, freq)
```

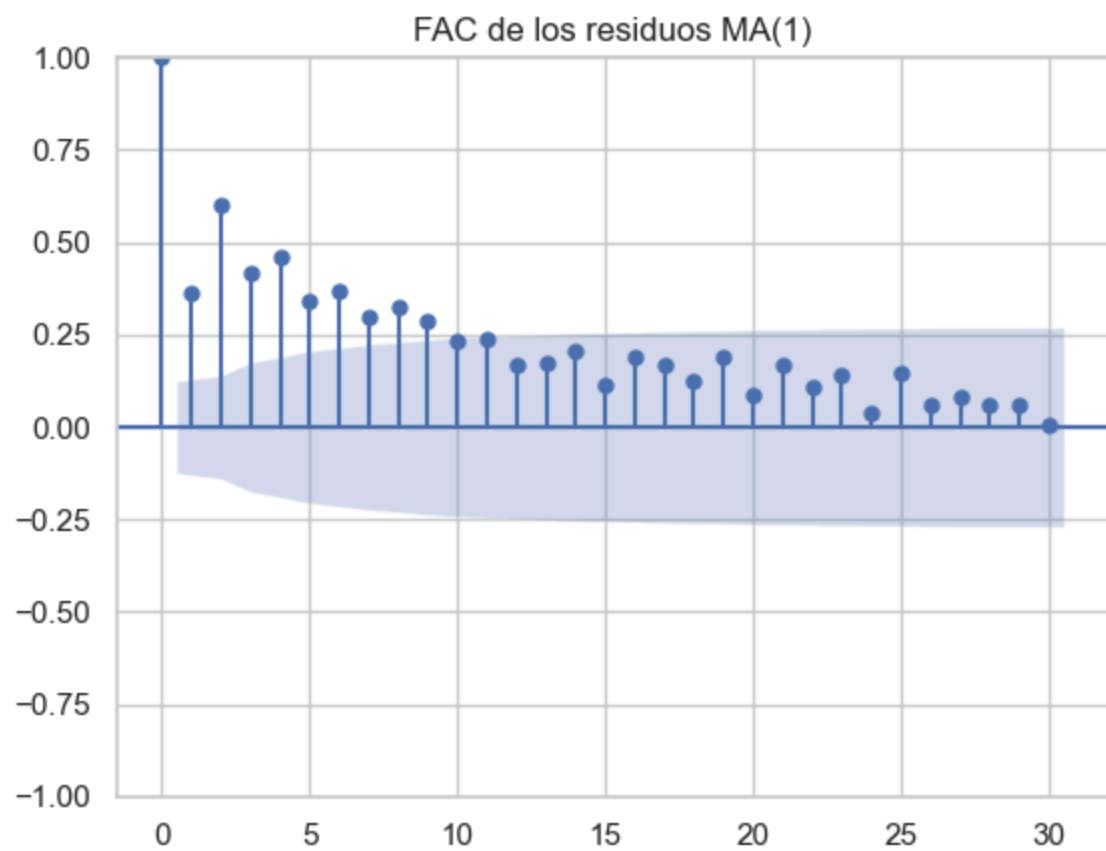
=== Prueba de Ljung-Box (residuos MA(1)) ===

```
      lb_stat      lb_pvalue
10  373.149588  4.832342e-74
```

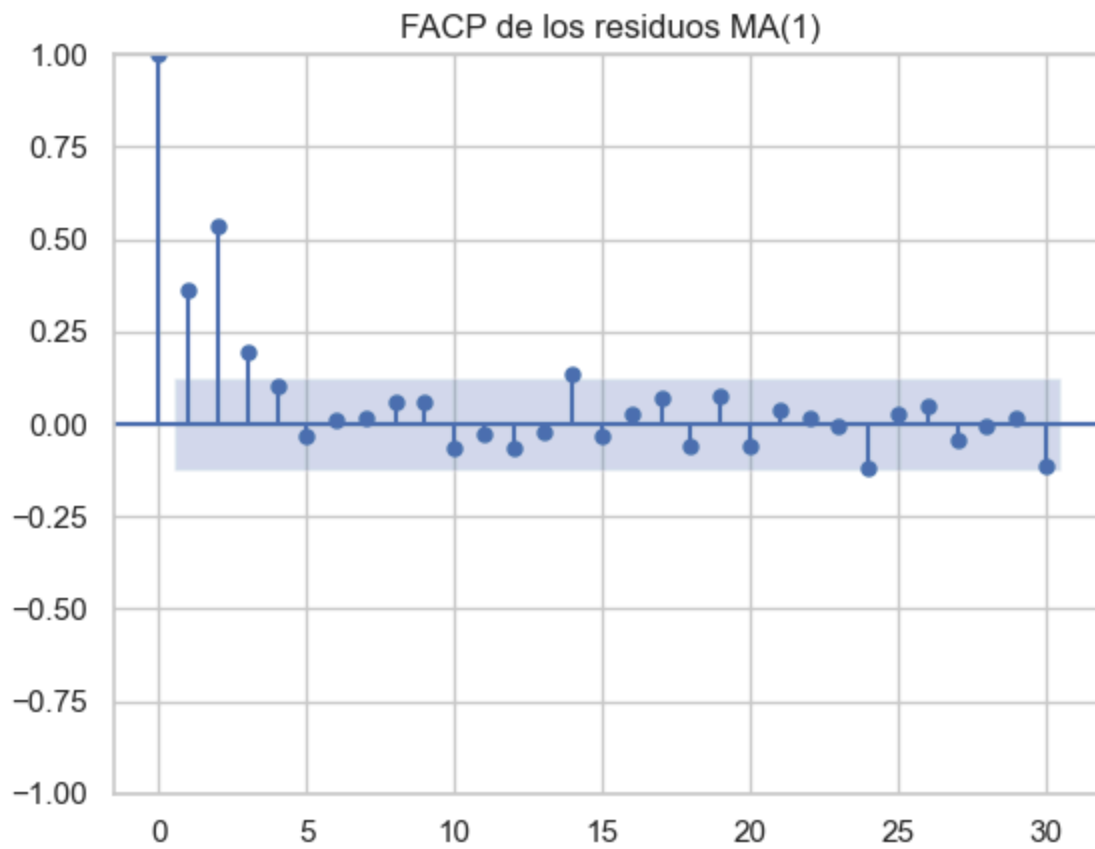
Conclusión: Se rechaza  $H_0$ ; los residuos muestran autocorrelación (modelo MA(1) puede ser insuficiente).



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<Figure size 800x400 with 0 Axes>



ARMA(1,1)

=== Modelo ARMA(1,1) estimado sobre la serie en niveles ===

#### SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(1, 0, 1)        Log Likelihood      -335.943
Date:             Wed, 19 Nov 2025      AIC                 679.885
Time:             22:57:59              BIC                 693.955
Sample:           01-01-2005            HQIC                685.549
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
const         43.4391      0.620      70.053      0.000      42.224      44.654
ar.L1          0.8998      0.039      23.089      0.000       0.823       0.976
ma.L1         -0.1875      0.053      -3.560      0.000      -0.291      -0.084
sigma2         0.8651      0.027      31.734      0.000       0.812       0.919
=====
Ljung-Box (L1) (Q):            0.16    Jarque-Bera (JB):            41584.32
Prob(Q):                      0.69    Prob(JB):                  0.00
Heteroskedasticity (H):        4.28    Skew:                      -5.50
Prob(H) (two-sided):          0.00    Kurtosis:                   65.35
=====
```

#### Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

#### Parámetros principales:

Constante: 43.4391

Phi\_1 (AR1): 0.8998

Theta\_1 (MA1): -0.1875

AIC: 679.89

BIC: 693.96

c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa\_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.

self.\_init\_dates(dates, freq)

c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa\_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.

self.\_init\_dates(dates, freq)

c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa\_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.

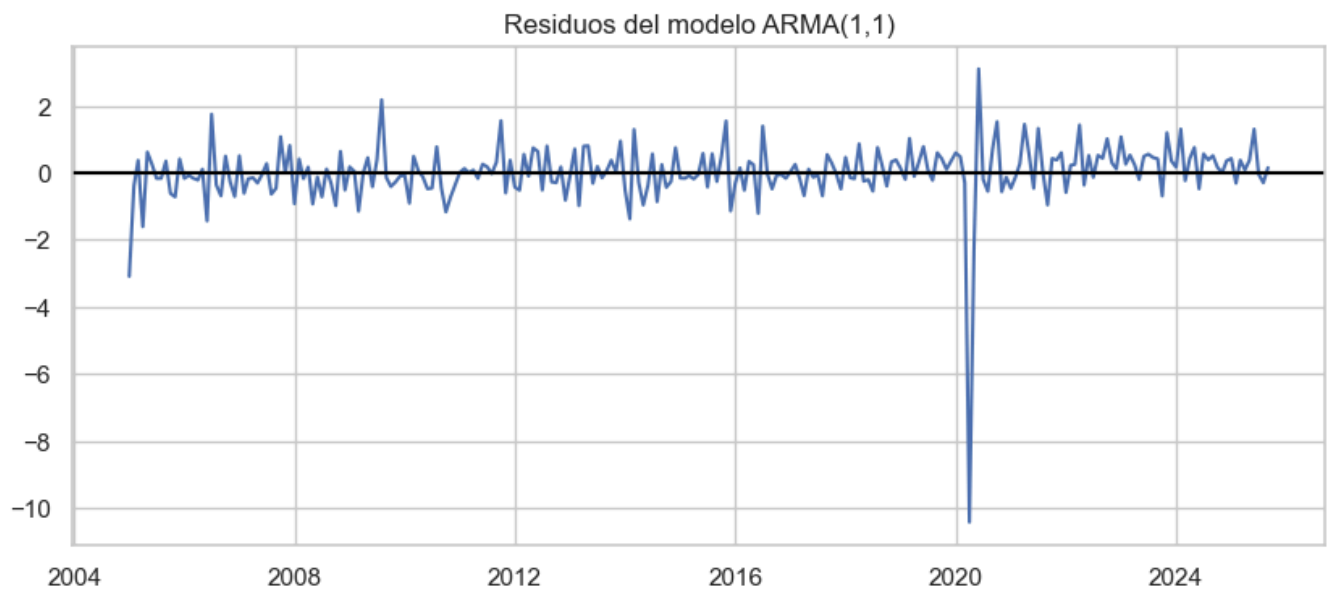
self.\_init\_dates(dates, freq)

=== Prueba de Ljung-Box (residuos ARMA(1,1)) ===

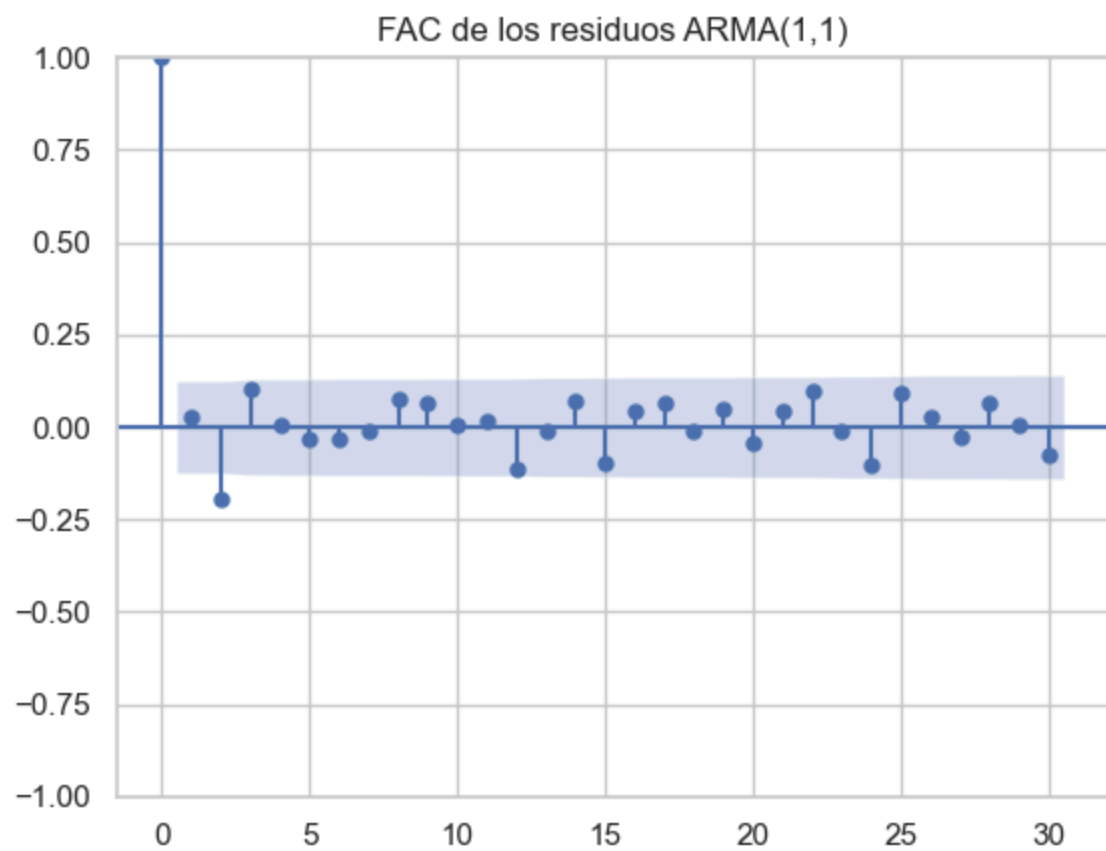
lb\_stat lb\_pvalue

10 15.931045 0.101623

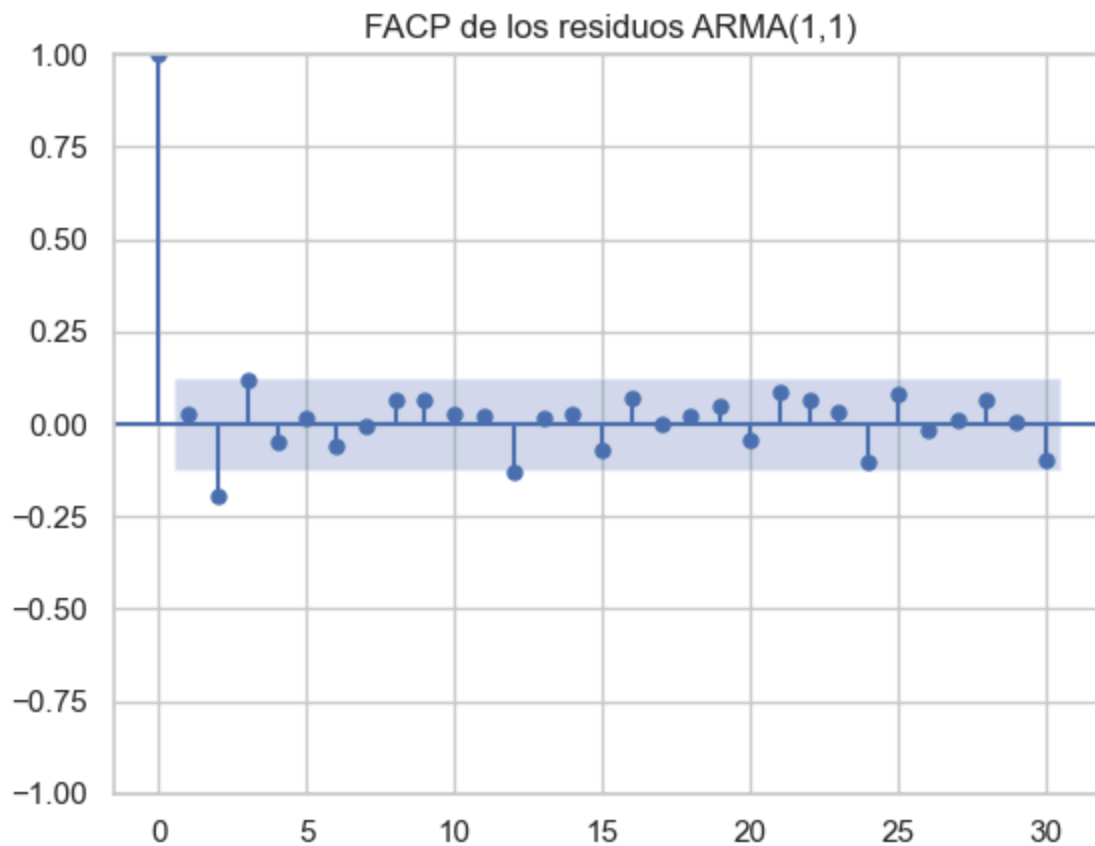
Conclusión: No se rechaza  $H_0$  de 'no autocorrelación' en los residuos (ruido blanco).



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Comparación de AIC

|   | Modelo    | AIC        | BIC        |
|---|-----------|------------|------------|
| 0 | AR(1)     | 681.295796 | 691.848155 |
| 1 | MA(1)     | 805.658408 | 816.210766 |
| 2 | ARMA(1,1) | 679.885451 | 693.955262 |

|   | Modelo    | Parametro | Estimacion | p_value       | Significativo_5pct |
|---|-----------|-----------|------------|---------------|--------------------|
| 0 | AR(1)     | const     | 43.449419  | 0.000000e+00  | True               |
| 1 | AR(1)     | ar.L1     | 0.848004   | 3.495771e-164 | True               |
| 2 | MA(1)     | const     | 43.472801  | 0.000000e+00  | True               |
| 3 | MA(1)     | ma.L1     | 0.745530   | 2.542017e-83  | True               |
| 4 | ARMA(1,1) | const     | 43.439123  | 0.000000e+00  | True               |
| 5 | ARMA(1,1) | ar.L1     | 0.899776   | 5.984189e-118 | True               |
| 6 | ARMA(1,1) | ma.L1     | -0.187464  | 3.701498e-04  | True               |

# Selección modelo final

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```

=== Modelo final ARMA(1,1) estimado sobre la serie de tasa de participación femenina ===

## SARIMAX Results

```
=====
Dep. Variable:      Tasa_participacion      No. Observations:      249
Model:              ARIMA(1, 0, 1)          Log Likelihood          -335.943
Date:              Wed, 19 Nov 2025         AIC                     679.885
Time:              22:58:00                 BIC                     693.955
Sample:            01-01-2005               HQIC                    685.549
                  - 09-01-2025
Covariance Type:    opg
=====
```

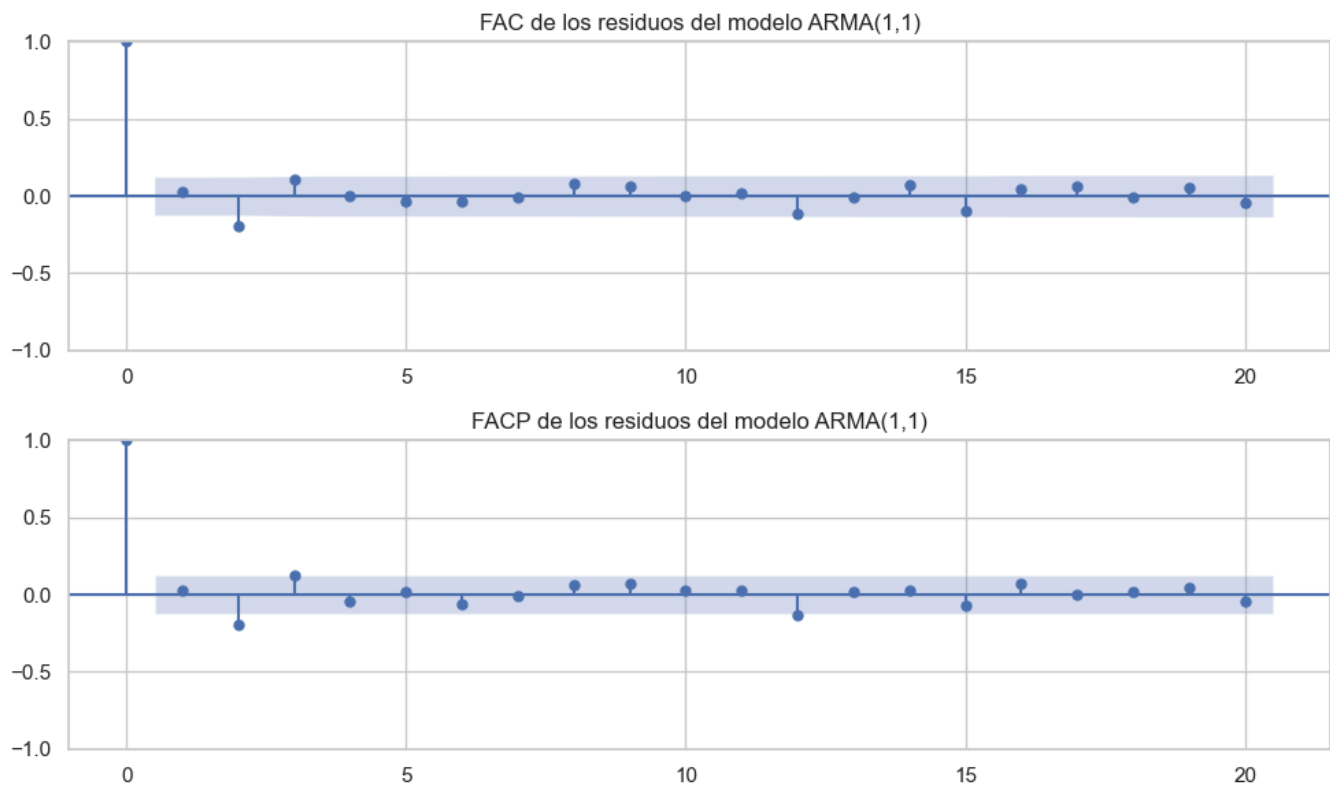
|        | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|--------|---------|---------|--------|-------|--------|--------|
| const  | 43.4391 | 0.620   | 70.053 | 0.000 | 42.224 | 44.654 |
| ar.L1  | 0.8998  | 0.039   | 23.089 | 0.000 | 0.823  | 0.976  |
| ma.L1  | -0.1875 | 0.053   | -3.560 | 0.000 | -0.291 | -0.084 |
| sigma2 | 0.8651  | 0.027   | 31.734 | 0.000 | 0.812  | 0.919  |

```
=====
Ljung-Box (L1) (Q):      0.16  Jarque-Bera (JB):      41584.32
Prob(Q):                 0.69  Prob(JB):              0.00
Heteroskedasticity (H):  4.28  Skew:              -5.50
Prob(H) (two-sided):     0.00  Kurtosis:          65.35
=====
```

## Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).





=== Modelo final ARMA(1,1) estimado sobre la serie de tasa de participación femenina ===  
SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(1, 0, 1)        Log Likelihood       -335.943
Date:             Wed, 19 Nov 2025      AIC                  679.885
Time:             22:58:00              BIC                  693.955
Sample:           01-01-2005            HQIC                 685.549
                  - 09-01-2025

Covariance Type:    opg
=====
```

|        | coef    | std err | z      | P> z  | [0.025 | 0.975] |
|--------|---------|---------|--------|-------|--------|--------|
| const  | 43.4391 | 0.620   | 70.053 | 0.000 | 42.224 | 44.654 |
| ar.L1  | 0.8998  | 0.039   | 23.089 | 0.000 | 0.823  | 0.976  |
| ma.L1  | -0.1875 | 0.053   | -3.560 | 0.000 | -0.291 | -0.084 |
| sigma2 | 0.8651  | 0.027   | 31.734 | 0.000 | 0.812  | 0.919  |

```
=====
Ljung-Box (L1) (Q):    0.16    Jarque-Bera (JB):    41584.32
Prob(Q):               0.69    Prob(JB):             0.00
Heteroskedasticity (H): 4.28    Skew:                -5.50
Prob(H) (two-sided):   0.00    Kurtosis:            65.35
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
self._init_dates(dates, freq)
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
self._init_dates(dates, freq)
```

## Pronóstico a 10 periodos con el ARMA(1,1)

=== Tabla de Pronóstico a 10 periodos ===

|            | Lead | Pronóstico | Error estándar | Límite inferior 95% \ |
|------------|------|------------|----------------|-----------------------|
| 2025-10-01 | 1    | 45.283087  | 0.930122       | 43.460083             |
| 2025-11-01 | 2    | 45.098278  | 1.141964       | 42.860070             |
| 2025-12-01 | 3    | 44.931992  | 1.288200       | 42.407167             |
| 2026-01-01 | 4    | 44.782371  | 1.395410       | 42.047416             |
| 2026-02-01 | 5    | 44.647745  | 1.476517       | 41.753826             |
| 2026-03-01 | 6    | 44.526613  | 1.539052       | 41.510126             |
| 2026-04-01 | 7    | 44.417620  | 1.587878       | 41.305437             |
| 2026-05-01 | 8    | 44.319552  | 1.626333       | 41.131997             |
| 2026-06-01 | 9    | 44.231312  | 1.656813       | 40.984018             |
| 2026-07-01 | 10   | 44.151916  | 1.681085       | 40.857050             |

|            | Límite superior 95% |
|------------|---------------------|
| 2025-10-01 | 47.106092           |
| 2025-11-01 | 47.336486           |
| 2025-12-01 | 47.456816           |
| 2026-01-01 | 47.517325           |
| 2026-02-01 | 47.541665           |
| 2026-03-01 | 47.543100           |
| 2026-04-01 | 47.529804           |
| 2026-05-01 | 47.507107           |
| 2026-06-01 | 47.478606           |
| 2026-07-01 | 47.446782           |



[illegible]

```
arning: Non-stationary starting autoregressive parameters found. Using zeros as starting parameters.
```

```
    warn('Non-stationary starting autoregressive parameters')
```

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\statespace\sarimax.py:978: UserWarning: Non-invertible starting MA parameters found. Using zeros as starting parameters.
```

```
    warn('Non-invertible starting MA parameters found.')
```

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
```

```
    self._init_dates(dates, freq)
```

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
```

```
    self._init_dates(dates, freq)
```

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency information was provided, so inferred frequency MS will be used.
```

```
    self._init_dates(dates, freq)
```

```
=== Modelos probados (búsqueda automática) ===
```

|   | order     | AIC        | BIC        |
|---|-----------|------------|------------|
| 0 | (1, 0, 2) | 670.936101 | 688.523366 |
| 1 | (1, 0, 1) | 679.885451 | 693.955262 |
| 2 | (1, 0, 0) | 681.295796 | 691.848155 |
| 3 | (2, 0, 0) | 681.620589 | 695.690400 |
| 4 | (2, 0, 1) | 682.231852 | 699.819117 |
| 5 | (0, 0, 2) | 764.521597 | 778.591409 |
| 6 | (0, 0, 1) | 805.658408 | 816.210766 |

=== Modelo automático elegido (por AIC) === ARIMA(1, 0, 2)

#### SARIMAX Results

```
=====
Dep. Variable:    Tasa_participacion    No. Observations:    249
Model:            ARIMA(1, 0, 2)        Log Likelihood       -330.468
Date:             Wed, 19 Nov 2025      AIC                  670.936
Time:             22:58:01              BIC                  688.523
Sample:           01-01-2005            HQIC                 678.015
                  - 09-01-2025
=====
```

Covariance Type: opg

```
=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
const         43.4320     0.755     57.543     0.000     41.953     44.911
ar.L1          0.9448     0.039     24.423     0.000      0.869      1.021
ma.L1         -0.1496     0.056     -2.665     0.008     -0.260     -0.040
ma.L2         -0.2506     0.075     -3.337     0.001     -0.398     -0.103
sigma2         0.8274     0.026     31.258     0.000      0.775      0.879
=====
```

```
=====
Ljung-Box (L1) (Q):           0.21    Jarque-Bera (JB):           47848.09
Prob(Q):                     0.65    Prob(JB):                   0.00
Heteroskedasticity (H):       3.76    Skew:                      -5.88
Prob(H) (two-sided):          0.00    Kurtosis:                   69.88
=====
```

#### Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

=== Pronóstico automático a 10 periodos ===

```
Pronóstico_auto
2025-10-01    45.432489
2025-11-01    45.307927
2025-12-01    45.204341
2026-01-01    45.106475
2026-02-01    45.014013
2026-03-01    44.926657
2026-04-01    44.844125
2026-05-01    44.766150
2026-06-01    44.692480
2026-07-01    44.622879
```

=== Comparación de pronósticos: modelo manual vs 'automático' ===

```
Pronóstico_manual_ARMA(1,1)  Pronóstico_auto_ARIMA(1, 0, 2)
2025-10-01    45.283087    45.432489
2025-11-01    45.098278    45.307927
2025-12-01    44.931992    45.204341
2026-01-01    44.782371    45.106475
2026-02-01    44.647745    45.014013
2026-03-01    44.526613    44.926657
2026-04-01    44.417620    44.844125
2026-05-01    44.319552    44.766150
2026-06-01    44.231312    44.692480
2026-07-01    44.151916    44.622879
```

# Serie actualizada

Nuevo dato observado (simulado): 45.2830873262225  
Última fecha histórica: 2025-09-01 00:00:00  
Nueva fecha simulada: 2025-10-01 00:00:00

Últimas observaciones de la serie actualizada:  
Fecha

|            |           |
|------------|-----------|
| 2025-06-01 | 46.670945 |
| 2025-07-01 | 46.065398 |
| 2025-08-01 | 45.516931 |
| 2025-09-01 | 45.521386 |
| 2025-10-01 | 45.283087 |

Name: Tasa\_participacion, dtype: float64

=== Modelo ARMA(1,1) reestimado con la serie actualizada ===

## SARIMAX Results

```
=====
Dep. Variable:      Tasa_participacion      No. Observations:      250
Model:              ARIMA(1, 0, 1)          Log Likelihood          -336.788
Date:               Wed, 19 Nov 2025        AIC                      681.576
Time:               22:58:01                BIC                      695.662
Sample:             01-01-2005              HQIC                   687.246
                  - 10-01-2025
=====
```

Covariance Type: opg

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
const         43.4391      0.618       70.320      0.000      42.228      44.650
ar.L1          0.8998      0.039       23.185      0.000       0.824       0.976
ma.L1         -0.1875      0.052       -3.575      0.000      -0.290      -0.085
sigma2         0.8617      0.027       31.864      0.000       0.809       0.915
=====
Ljung-Box (L1) (Q):              0.16      Jarque-Bera (JB):              42096.37
Prob(Q):                        0.69      Prob(JB):                  0.00
Heteroskedasticity (H):          4.28      Skew:                      -5.51
Prob(H) (two-sided):            0.00      Kurtosis:                  65.61
=====
```

## Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

```
c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
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c:\Users\narro\miniconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarni
ng: No frequency information was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```

=== Tabla de Pronóstico ACTUALIZADO a 9 periodos ===

|            | Lead | Pronóstico actualizado | Error estándar | Límite inferior 95% \ |
|------------|------|------------------------|----------------|-----------------------|
| 2025-11-01 | 1    | 45.098320              | 0.928257       | 43.278969             |
| 2025-12-01 | 2    | 44.932059              | 1.139676       | 42.698335             |
| 2026-01-01 | 3    | 44.782459              | 1.285625       | 42.262680             |
| 2026-02-01 | 4    | 44.647849              | 1.392629       | 41.918347             |
| 2026-03-01 | 5    | 44.526728              | 1.473581       | 41.638562             |
| 2026-04-01 | 6    | 44.417745              | 1.536000       | 41.407240             |
| 2026-05-01 | 7    | 44.319682              | 1.584736       | 41.213656             |
| 2026-06-01 | 8    | 44.231446              | 1.623123       | 41.050183             |
| 2026-07-01 | 9    | 44.152051              | 1.653549       | 40.911155             |

|            | Límite superior 95% |
|------------|---------------------|
| 2025-11-01 | 46.917671           |
| 2025-12-01 | 47.165783           |
| 2026-01-01 | 47.302237           |
| 2026-02-01 | 47.377351           |
| 2026-03-01 | 47.414894           |
| 2026-04-01 | 47.428250           |
| 2026-05-01 | 47.425708           |
| 2026-06-01 | 47.412708           |
| 2026-07-01 | 47.392948           |

